GALAXY EVOLUTION

Machine Learning for Stellar Mass
and Star Formation Rate Prediction

SCIENTIFIC PROJECT REPORT

Dataset: KiDS DR4 galaxy catalogue
Targets: catalogue stellar-mass and SFR estimates
Report date: 8 November 2026

Abstract

This project develops supervised machine-learning regression models to predict two galaxy properties from photometric information in the KiDS DR4 galaxy catalogue: the catalogue stellar-mass estimate (MASS_MED_GaZNet) and the catalogue star-formation-rate estimate (SFR_MED_GaZNet). The modelling dataset contains 288,809 records. Nineteen inputs were selected: nine GAaP magnitudes, their nine photometric errors, and GaZNet photometric redshift. Separate Ridge Regression and HistGradientBoostingRegressor models were trained for each target using a redshift-stratified 60/20/20 split. HistGradientBoosting performed best on validation data. Test performance was MAE 0.099398, RMSE 0.140408, R^2 0.955023 for stellar mass, and MAE 0.650390, RMSE 1.588423, R^2 0.548427 for SFR. The SFR model performed substantially worse on 240 extreme-low records below –10. These models reproduce catalogue-derived estimates and are not independently validated against physical ground truth.

Keywords

Galaxy evolution; KiDS DR4; regression; stellar mass; star formation rate; photometry; redshift.

1. Introduction

Galaxy evolution is studied through properties such as stellar mass and star formation rate (SFR). Large photometric surveys provide measurements across multiple wavelength bands. This project uses supervised regression to learn relationships between photometric measurements, a photometric redshift estimate, and catalogue-derived target estimates.

1.1 Objectives

- Prepare a modelling table without modifying the original catalogue.
- Predict MASS_MED_GaZNet and SFR_MED_GaZNet from 19 photometric/redshift inputs.
- Compare Ridge Regression and HistGradientBoosting using validation metrics.
- Evaluate the selected models on an untouched test set.
- Study feature importance, subgroup and redshift performance, and residuals.
- Save models and test predictions for reproducibility.

2. Dataset and Variables

The compressed FITS source file was KiDS_DR4_galaxy_catalog.fits.gz, read from HDU 1. The joined catalogue contained 288,809 rows and 135 columns. The selected 19 predictors and two targets had no missing values.

Item	Description
Dataset	KiDS DR4 galaxy catalogue (FITS, compressed)
Catalogue records	288,809 rows
Catalogue columns	135 columns
Selected input features	19 (9 GAaP magnitudes, 9 corresponding errors, and GaZNet redshift)
Prediction targets	MASS_MED_GaZNet and SFR_MED_GaZNet
Selected-feature missingness	No missing values in the 19 inputs or 2 targets

Inputs: MAG_GAAP_u, MAG_GAAP_g, MAG_GAAP_r, MAG_GAAP_i, MAG_GAAP_Z, MAG_GAAP_Y, MAG_GAAP_J, MAG_GAAP_H, MAG_GAAP_Ks, MAGERR_GAAP_u, MAGERR_GAAP_g, MAGERR_GAAP_r, MAGERR_GAAP_i, MAGERR_GAAP_Z, MAGERR_GAAP_Y, MAGERR_GAAP_J, MAGERR_GAAP_H, MAGERR_GAAP_Ks, GaZNet. Targets: MASS_MED_GaZNet and SFR_MED_GaZNet.

The target columns are catalogue estimates. Their physical units and exact logarithmic conventions were not confirmed in the recorded analysis, so no units are assigned to the reported errors.

3. Data Audit and Preparation

The SFR target contained a small extreme-low tail: 1,234 full-catalogue records (about 0.427%) had values below –10. This was used as a descriptive subgroup boundary, not as a documented invalid-value code. These records were retained. Photometric-error distributions were inspected; large maxima were associated with faint sources, so high-error records were not automatically removed. Per-band 99th-percentile flags were retained as quality indicators.

Subset	Fraction	Records
Training	60%	173,285
Validation	20%	57,762
Test	20%	57,762

The split was stratified by GaZNet deciles using qcut (10 quantile bins, duplicate edges dropped) and random_state=42. It produced 173,285 training, 57,762 validation, and 57,762 test records.

4. Modelling Methodology

4.1 Algorithms

Ridge Regression used StandardScaler followed by Ridge(alpha=1.0). HistGradientBoostingRegressor used max_iter=150, learning_rate=0.08, max_leaf_nodes=31, l2_regularization=1.0, and random_state=42. Independent estimators were fitted for each target.

4.2 Metrics

- MAE: mean absolute prediction error.
- RMSE: square root of mean squared error; larger errors receive greater weight.
- R^2: proportion of target variance explained relative to a constant-mean baseline; negative values indicate worse performance than that baseline on the evaluated subset.

4.3 Validation model comparison

Target	Model	MAE	RMSE	R^2
Stellar mass	Ridge Regression	0.128541	0.183105	0.924130
Stellar mass	HistGradientBoosting	0.099695	0.141703	0.954561
SFR	Ridge Regression	0.834601	1.640037	0.469739
SFR	HistGradientBoosting	0.645245	1.479176	0.568658

HistGradientBoosting had lower MAE and RMSE and higher R^2 than Ridge Regression for both targets, so it was selected for final testing.

5. Feature Importance

Permutation importance was computed on a reproducible 10,000-record validation sample, with three repeats and negative MAE scoring. A feature's importance is the increase in MAE after shuffling it. It is model-dependent and does not establish causality.

5.1 Stellar mass

The leading features were GaZNet (0.417820), MAG_GAAP_Z (0.305427), MAG_GAAP_Ks (0.164455), MAGERR_GAAP_u (0.070490), MAG_GAAP_J (0.058215), and MAGERR_GAAP_g (0.051920).

Feature	Importance
GaZNet	0.417820
MAG_GAAP_Z	0.305427
MAG_GAAP_Ks	0.164455
MAGERR_GAAP_u	0.070490
MAG_GAAP_J	0.058215
MAGERR_GAAP_g	0.051920
MAGERR_GAAP_r	0.024803
MAG_GAAP_Y	0.015649
MAG_GAAP_g	0.010147
MAGERR_GAAP_Ks	0.009883

Feature	Importance
MAG_GAAP_i	0.008881
MAG_GAAP_r	0.007905
MAG_GAAP_H	0.006283
MAG_GAAP_u	0.005590
MAGERR_GAAP_H	0.000088
MAGERR_GAAP_J	0.000063
MAGERR_GAAP_i	0.000032
MAGERR_GAAP_Z	0.000028
MAGERR_GAAP_Y	−0.000001

5.2 SFR

The leading features were MAG_GAAP_Z (4.804347), MAG_GAAP_Ks (3.851918), MAG_GAAP_u (3.677025), MAG_GAAP_Y (3.044810), MAG_GAAP_H (2.269297), and MAG_GAAP_r (1.479475). GaZNet importance was 0.347579. Correlated bands can share predictive information.

Feature	Importance
MAG_GAAP_Z	4.804347
MAG_GAAP_Ks	3.851918
MAG_GAAP_u	3.677025
MAG_GAAP_Y	3.044810
MAG_GAAP_H	2.269297
MAG_GAAP_r	1.479475
MAG_GAAP_i	0.680465
MAG_GAAP_J	0.571000
GaZNet	0.347579
MAGERR_GAAP_r	0.323153
MAG_GAAP_g	0.295705
MAGERR_GAAP_Z	0.122290
MAGERR_GAAP_u	0.105253
MAGERR_GAAP_g	0.073373
MAGERR_GAAP_Y	0.056696
MAGERR_GAAP_H	0.049254
MAGERR_GAAP_Ks	0.025545
MAGERR_GAAP_i	0.021843
MAGERR_GAAP_J	0.006378

6. Final Test-Set Results

Target	Test MAE	Test RMSE	Test R^2
Stellar mass	0.099398	0.140408	0.955023
SFR	0.650390	1.588423	0.548427

6.1 Stellar mass

The stellar-mass model achieved MAE 0.099398, RMSE 0.140408, and R^2 0.955023. Most test points were close to the identity line, indicating strong agreement with the catalogue target.

6.2 SFR

The SFR model achieved MAE 0.650390, RMSE 1.588423, and R^2 0.548427. The plot shows the main ordinary-value population but substantial compression of the most negative actual values.

6.3 SFR subgroup performance

Group	N	MAE	RMSE	R^2
Ordinary SFR (≥ –10)	57,522	0.588146	0.971130	0.698189
Extreme SFR (< –10)	240	15.568609	19.524540	–1.284019

The ordinary group (target ≥ –10) had R^2 0.698189. The extreme group (target < –10) had R^2 –1.284019 and MAE 15.568609. The extreme records were retained because they were not established as invalid.

6.4 Performance by redshift

GaZNet	N	Mass MAE	Mass R^2	SFR MAE	SFR R^2
0–0.3	39,228	0.104083	0.950451	0.599994	0.524476
0.3–0.5	13,583	0.084720	0.892182	0.736823	0.577971
0.5–1	4,887	0.100541	0.816703	0.806868	0.642051
1–3	64	0.255846	0.582223	1.246830	–60.610406

The highest interval (1–3) contained only 64 test records. Its SFR R^2 of –60.610406 is unstable and should not be generalized. MAE increased in the highest interval for both targets.

7. Residual Analysis

Residual = actual – predicted. Positive values indicate underprediction; negative values indicate overprediction.

Subset	Mean	Median	SD	Max |error|
Stellar mass	0.000518	–0.003833	0.140407	2.131851
SFR – all	0.007114	0.060595	1.588407	53.608432
SFR – ordinary	0.070743	0.062336	0.968550	27.128988
SFR – extreme	–15.243203	–8.396988	12.200510	53.608432

Stellar-mass mean and median residuals (0.000518 and –0.003833) were close to zero. For SFR, the ordinary-group mean residual was +0.070743, while the extreme-group mean residual was –15.243203. The two groups partially cancel in the overall mean residual (+0.007114), so the aggregate value hides subgroup bias.

8. Diagnostic Figures

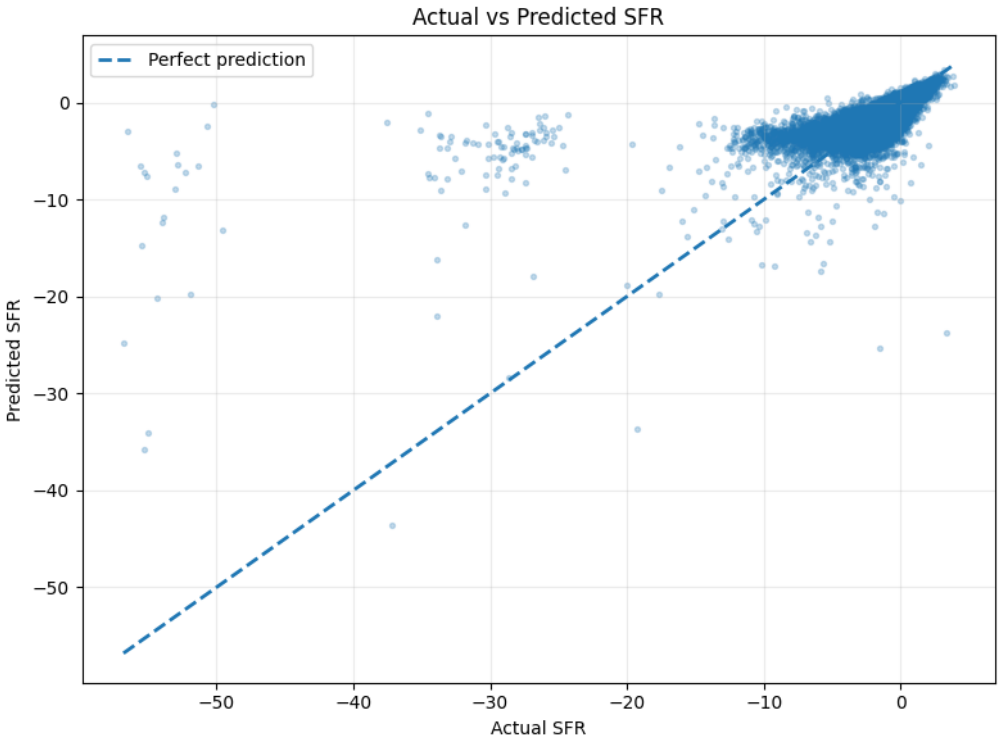


Figure 1. Actual versus predicted stellar mass on the held-out test set.

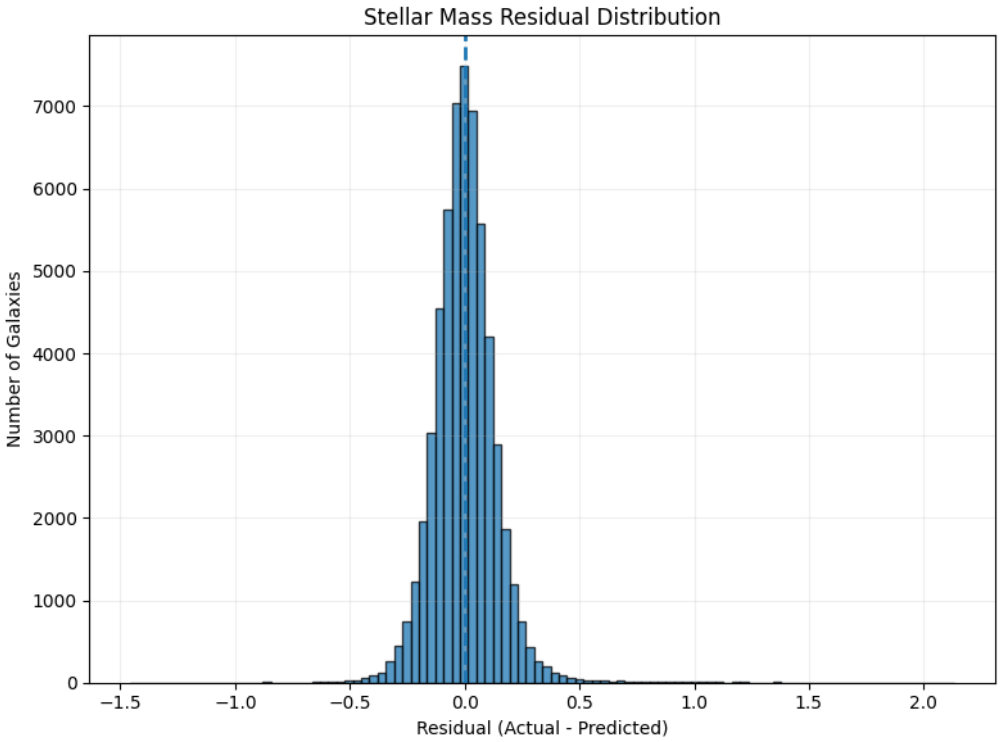


Figure 2. Actual versus predicted SFR on the held-out test set.

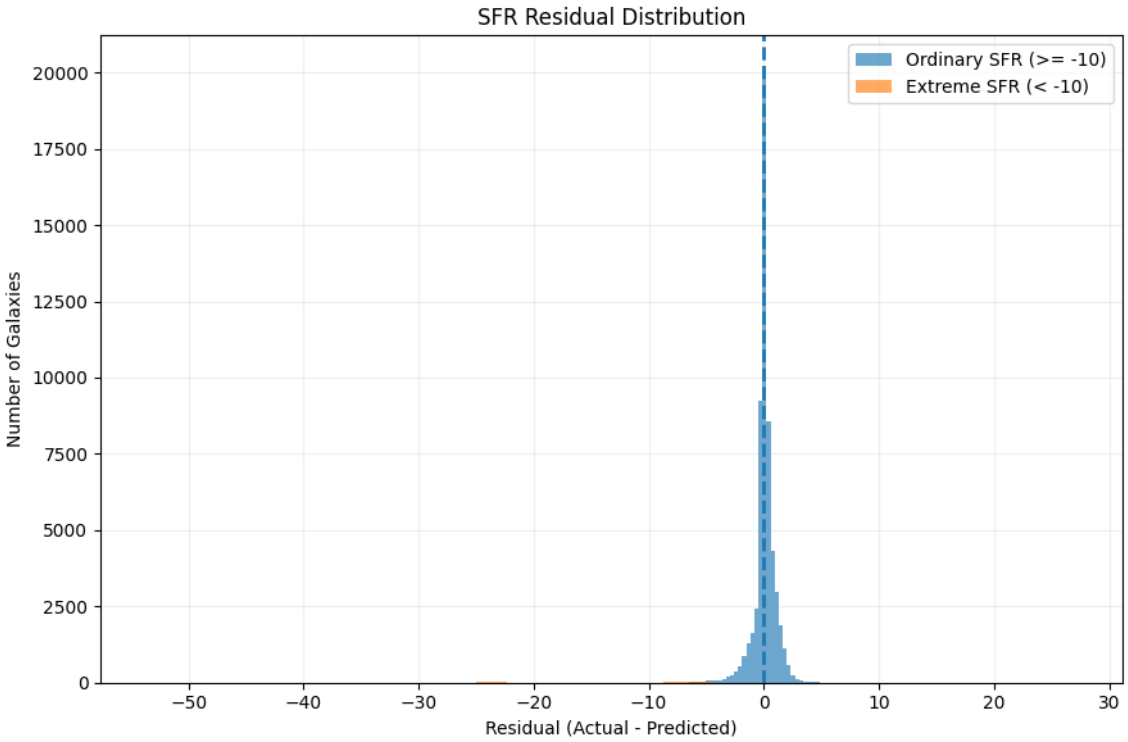


Figure 3. Stellar-mass residual distribution (residual = actual – predicted).

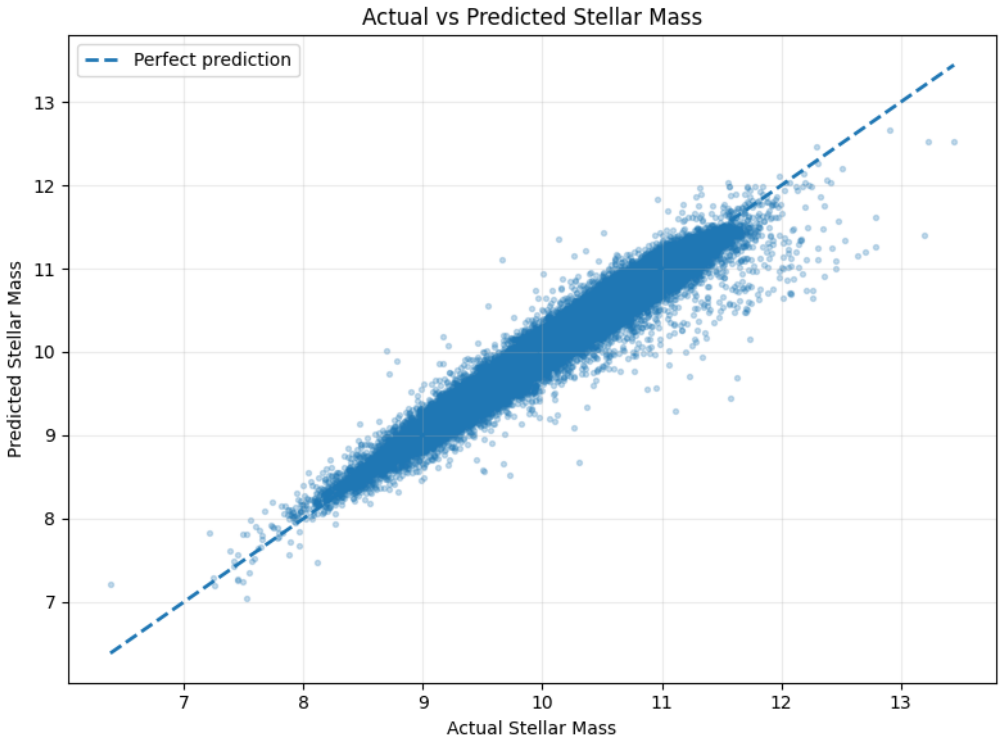


Figure 4. SFR residual distribution, showing ordinary and extreme groups.

9. Discussion

The stellar-mass model reproduces its catalogue target strongly, while SFR prediction is moderate overall and weak for the extreme-low subgroup. Validation and test metrics are closely aligned: stellar-mass validation/test MAE were 0.099695/0.099398 and R^2 were 0.954561/0.955023; SFR validation/test MAE were 0.645245/0.650390 and R^2 were 0.568658/0.548427. This consistency is encouraging and does not show a substantial validation-to-test performance gap.

10. Limitations and Scientific Cautions

- The targets are GaZNet-associated catalogue estimates, not independent physical ground truth.

- GaZNet is an input and is associated with the target-estimation process; this creates possible circularity relative to independent physical inference.
- Target units and exact logarithmic conventions were not verified; metrics are in the native numerical scales of the target columns.
- The SFR < –10 boundary is descriptive, not a documented invalid-value threshold.
- Only 64 test records occupy the GaZNet 1–3 interval, making its subgroup metrics unstable.
- Permutation importance is model- and sample-dependent; correlated inputs can share importance.
- The held-out test set should not be used for further model tuning.

11. Conclusions

A supervised workflow was developed for predicting two catalogue-derived galaxy properties from 19 features in 288,809 KiDS DR4 records. HistGradientBoosting outperformed Ridge Regression on validation data. On the held-out test set, stellar mass achieved MAE 0.099398, RMSE 0.140408, R^2 0.955023; SFR achieved MAE 0.650390, RMSE 1.588423, R^2 0.548427. The SFR model performed considerably better for ordinary records than for the extreme-low tail. These findings support the stellar-mass model as a strong surrogate for the catalogue target and the SFR model as a more limited surrogate, subject to the stated caveats.

11.1 Future work

- Verify target units and conventions from authoritative catalogue documentation.
- Investigate extreme-low SFR records using source-level quality metadata before applying any special treatment.
- Test robust or group-aware SFR models using training and validation data only.
- Use grouped permutation or feature-ablation analysis for correlated photometric bands.
- Validate against independent physical reference measurements for independent physical inference.
- Increase high-redshift sample sizes or report uncertainty intervals.

12. Reproducibility and Saved Artifacts

Folder	Artifact
04_trained_models	galaxy_evolution_hgb_models.joblib
04_trained_models	galaxy_evolution_test_predictions.csv
05_graphs	01_stellar_mass_prediction.png
05_graphs	02_sfr_prediction.png
05_graphs	03_stellar_mass_residuals.png
05_graphs	04_sfr_residuals_by_group.png

The model bundle contains the separate fitted estimators, feature names, target names, and split notes. The test-predictions CSV contains redshift, actual and predicted targets, residuals, and SFR subgroup labels. The report should be stored in 06_results.

Appendix A. Model configuration

Ridge: StandardScaler + Ridge(alpha=1.0). HistGradientBoostingRegressor: max_iter=150, learning_rate=0.08, max_leaf_nodes=31, l2_regularization=1.0, random_state=42. Split: 60%/20%/20%, stratified by GaZNet deciles, random_state=42.